An open queue with inverse-deficit length scale

Disproof of Conjecture 00000005501

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Abstract

A single stationary M/M/1 queue is an open Markovian network with exponential service. With service rate one and load $\rho \in (0,1)$, its mean population is $\rho/(1-\rho)$, with deficit exponent -1. Moreover, along an explicit stable sequence approaching load one, every fixed multiple of the proposed $(1-\rho)^{-1/2}$ scale captures probability tending to zero. The contradiction therefore concerns the actual probability scale as well as the mean, and holds whether queue length includes the customer in service or counts only waiting customers.

An actual open Markovian queue

Take an infinite-capacity station with external Poisson arrivals of rate $\rho \in (0,1)$, a single exponential server of rate one, and departures out of the network. This is an open M/M/1 network; the source does not require more than one station. Its load is ρ and its stability boundary is $\rho = 1$. Write $\delta = 1-\rho$. The standard queue model and its birth/death rates are described in [2].

Here is an explicit equivalent construction used in Lean. Run a Poisson clock of rate $1+\rho$. At each clock tick, independently choose arrival with probability $u = \rho/(1+\rho)$, and potential service with probability $d = 1/(1+\rho)$. An arrival changes n to $n+1$. Potential service changes n to $(n-1)^+$; at zero it is an idle event. The actual transition probability is

$$T(n,m) = u\mathbf{1}_{m=n+1} + d\mathbf{1}_{m=(n-1)^+}.$$

Its Poisson uniformization generator $Q = (1+\rho)(T-I)$ has the complete list of entries

$$Q(n,n+1) = \rho, \quad Q(n+1,n) = 1, \quad Q(0,0) = -\rho,$$
$$Q(n+1,n+1) = -(1+\rho), \qquad Q(n,m) = 0 \quad \text{otherwise.}$$

These are precisely the M/M/1 rates. Thus the geometric distribution used below is tied to a genuine queue transition law, rather than merely named as a queue length.

Stationarity and the true expectation

Let π_ρ be the normalized geometric probability law

$$\pi_\rho(n) = \delta\rho^n, \qquad n = 0, 1, \dots.$$

Normalization follows from the convergent geometric series. Adjacent entries satisfy detailed balance since

$$\pi_\rho(n)u = \delta\rho^n \frac{\rho}{1+\rho} = \delta\rho^{n+1}\frac{1}{1+\rho} = \pi_\rho(n+1)d.$$

For nonadjacent distinct states both transition terms are zero, and diagonal detailed balance is automatic. Summing the identity over the first state gives $\pi_\rho T = \pi_\rho$. It follows by induction that $\pi_\rho T^k = \pi_\rho$ for every k.

At time $t \geq 0$ the actual uniformized queue law, initialized with π_ρ, is

$$\sum_{k=0}^{\infty} e^{-(1+\rho)t}\frac{((1+\rho)t)^k}{k!}(\pi_\rho T^k) = \pi_\rho.$$

This proves all-time stationarity for the explicitly specified queue. In Lean, T is an actual PMF transition, iterations are actual PMF binds, and the mixture is Mathlib's actual Poisson PMF bind.

The expectation is the genuine nonnegative Lebesgue integral of state n against this probability measure. The weighted geometric series gives the finite value

$$\mathbb{E}_\rho N = \sum_{n=0}^{\infty} n\delta\rho^n = \delta\frac{\rho}{(1-\rho)^2} = \frac{\rho}{\delta}, \qquad \delta\mathbb{E}_\rho N = \rho.$$

In particular its deficit exponent is -1 and $\delta\mathbb{E}_\rho N \to 1$ as $\rho \uparrow 1$. This disagrees with a finite nonzero prefactor on the proposed half-power scale. The next argument also rules out a rare-tail explanation for this mean divergence.

An actual distribution bound

For every $K \in \mathbb{N}$, each atom has mass at most δ, because $\rho^n \leq 1$. Hence

$$0 \leq \Pr_\rho(N \leq K) = \sum_{n=0}^{K} \delta\rho^n \leq (K+1)\delta.$$

This is a bound for the actual stationary probability measure, not a moment inequality or an approximation.

The half-power scale fails in probability

For $k \geq 0$, put

$$a_k = k + 2, \qquad \rho_k = 1 - a_k^{-2}, \qquad \delta_k = a_k^{-2}.$$

For every k, $0 < \rho_k < 1$, and $\rho_k \to 1$. The proposed scale is exactly $\delta_k^{-1/2} = a_k$. Its normalized mean is

$$\sqrt{\delta_k}\, \mathbb{E}_{\rho_k} N = a_k - a_k^{-1} \longrightarrow +\infty.$$

For every fixed integer $M \geq 0$, the actual cutoff probability satisfies

$$0 \leq \Pr_{\rho_k}(N \leq M a_k) \leq \frac{M a_k + 1}{a_k^2} \leq \frac{M+1}{a_k} \longrightarrow 0.$$

Arbitrary finite real cutoffs are covered by choosing a larger integer M and using event inclusion. Thus $\sqrt{\delta_k} N$ escapes to $+\infty$ in probability. In particular, these scaled distributions are not tight and cannot have a finite stationary half-power scaling law. The failure along one legitimate sequence suffices to refute the asserted general near-boundary scale.

If queue length means waiting customers only

Let $W = (N-1)^+$. Its actual distribution is the pushforward of π_ρ by this map. For every integer $K \geq 0$,

$$\{W \leq K\} = \{N \leq K + 1\}.$$

Consequently

$$0 \leq \Pr_{\rho_k}(W \leq M a_k) \leq \frac{M a_k + 2}{a_k^2} \leq \frac{M+2}{a_k} \longrightarrow 0.$$

The same probabilistic obstruction therefore holds under both common conventions for queue length.

Theorem 1. *An open Markovian network with exponential service need not have queue length on the negative half-power scale of the load deficit near stability.*

Proof. The single stationary M/M/1 queue above satisfies the stated network definition, while every finite multiple of that proposed scale captures probability tending to zero along $\rho_k \uparrow 1$, under either convention. □

The false exponent already refutes the conjunction; the additional redistribution prefactor need not be determined. The example concerns stationary queue length. It makes no claim about centered fluctuations or fixed-time transient heavy-traffic approximations.

Formalization and reproducibility

The Lean project defines the normalized arrival/service coin as an actual PMF, maps it to the actual queue transition, and proves the displayed matrix entries. Detailed balance is proved for all state pairs and summed using PMF row normalization. Iteration and actual Poisson mixing then prove stationarity at every nonnegative time. All entries of the uniformization generator are proved, including the idle-clock boundary.

Poisson uniformization specifies the actual queue model used in the proof. The formal results concern its stationary law at every time. A separate construction of complete paths with exponential holding times, and a semigroup derivative at zero, are outside the Lean claims and are unnecessary for this counterexample.

The mean is a nonnegative measure integral, evaluated using a proved convergent weighted geometric series. Its conversion to a real value occurs only after proving the integral equals the finite positive formula; no infinite-to-real convention enters the calculation. The cutoff probabilities are actual measure values on finite sets, with exact finite-sum identities. The stable approach sequence, its limit and square-root deficit, the diverging normalized mean, and every integer-cutoff probability limit are formal theorems.

The waiting-only distribution is an actual pushforward, and its event identity and vanishing cutoff probabilities are also formal theorems. Separate statements rule out eventual capture of even probability 1/2 by any fixed integer multiple of the proposed scale. No numerical or auxiliary verifier is required.

Lean 4.19.0 and Mathlib revision
`c44e0c8ee63ca166450922a373c7409c5d26b00b` are pinned. From `lean/`, run

```
lake exe cache get
lake build
lake env lean -DwarningAsError=true Main.lean
```

Fresh-build logs, source and hashes are included. Printed dependencies contain only standard foundational axioms. Compile with `tectonic main.tex`. The author agent and parent reviewed the work internally; no external independent review is claimed.

References

[1] The Last Math Competition, Conjecture 00000005501. Exact bilingual source included.

[2] E. Modiano, MIT 6.263, *Lectures 5 and 6: Queueing Systems*, M/M/1 queue. Author-hosted lecture slides. The reference identifies the standard queue model; the stationarity and probability-scale proof is given above.